



END SEMESTER ASSESSMENT (ESA) - JULY - 2023

UE18CS353 - Object Oriented Analysis and Design with Software Engineering

Total Marks : 100.0

1.a. Briefly explain about Requirements Engineering Process with a neat diagram

(6.0 Marks)

1.b. Write a use case diagram for Online Shopping website, considering any e-commerce application.

(6.0 Marks)

1.c. Discuss the V Model of Software Development Lifecycle with the different phases, focus of each of these phases and the deliverables or artifacts which you would see at the end of the phase. Discuss how is this similar to a waterfall model and how is it different from a waterfall model.

(8.0 Marks)

2.a. Contrast Architectural Views, Styles and Patterns. Give an example for each

(6.0 Marks)

2.b. Define the following wrt the class diagram:

- a) Association
- b) Aggregation
- c) Composition
- d) Multiplicity
- e) Qualifier
- f) Generalization

(6.0 Marks)

2.c. Develop an activity diagram for login of an application. Identify atleast 4 activities, specify all the transitions. (8.0 Marks)

3.a. Explain SOLID design principles. Give an example for any 2 principles (8.0 Marks)

3.b. Define the following: Preventive Maintenance, Perfective Maintenance, Corrective Maintenance, Adaptive Maintenance (4.0 Marks)

3.c. Define the following wrt Quality of an Agile Scrum project:
i) Sprint Burndown
ii) Team Velocity Metric
iii) Throughput
iv) Cycle Time (4.0 Marks)

3.d. Enlist the benefits of using Software configuration System. (4.0 Marks)

4.a. Differentiate between Static Testing and Dynamic Testing (6.0 Marks)

4.b. Contrast between Manual and Automation testing	(6.0 Marks)
4.c. Briefly discuss about different levels of testing.	(8.0 Marks)
5.a. Discuss the following: 1: Hacking and Software engineering 2: Availability Management	(6.0 Marks)
5.b. Enlist the pillars of Devops and specify few factors about each pillar of Devops	(8.0 Marks)
5.c. Briefly discuss about ITSM and ITIL	(6.0 Marks)